

Climate and Vegetation Across the Region

Lesson 3.06.04 | Week 16 | The Great Map Room

SS.3.G.2.1 | SS.3.G.2.4 | SS.3.G.4.1

Thinking Move: Sort | Compare | Explain

THE CLIMATE AND VEGETATION PANELS

A long row of colorful panels stretches from one end of the gallery to the other.

Icy white at one end. Sandy desert in the middle.

Lush green palms at the other.

They're missing their labels.

CHAT FRAME

"Last time we sorted the landmark photos."

"Today, we step over to the new panels -- climate and vegetation across the region."

"You learned about climate at the world scale in Unit 5. Now we zoom in."

Courtesy

It's easy to call a desert "empty" or the Arctic "boring" -- but people live in every one of these climate zones.

Courtesy means describing each place fairly. The desert is not less than the rainforest. Each one supports a different kind of life.

YOUR ROLE

Regional Curator

Your job: help visitors see how the same region can hold many different kinds of land and weather. Today you'll read a climate map and match plants to their zones.

Why does the land in one part of North America and the Caribbean look and feel so different from another part?

TODAY'S OBJECTIVES

Name the main climate zones across the region.

Match plants and trees to the climate zones where they grow.

Explain how climate shapes the way a place looks and feels.

YOU'LL NEED

Curator's Notebook

Pencil

Lesson Page

What Is Climate?

Climate is the usual weather of a place over many, many years. Not "what's the weather today?" but "what is the weather *usually* like here?"

North America and the Caribbean is one region -- but it stretches a very long way from north to south. That's why it holds five different climate zones, and each one grows different plants.

SOCRATIC QUESTIONS

- What's the difference between weather and climate?
- Why would a region that stretches from north to south have many climates?
- If you moved far north, what would change about the weather pattern?

INSERT IMAGE

Simple diagram or icon set:
snowflake (north) to sun/palm (south)
showing the gradient of climate zones

The Cold North

ARCTIC TUNDRA

Far northern Canada

Freezing cold most of the year. The ground stays frozen so deep that trees can't grow there -- only small grasses, moss, and tough little shrubs.

Key Plant: Arctic Moss

A tiny low-growing plant that survives on rocks and frozen ground where trees can't grow.

BOREAL FOREST

Most of Canada and northern U.S.

Long, snowy winters. Tall evergreen trees like pine and spruce grow here because they can survive cold winters by keeping their needles all year.

Key Plant: Spruce Tree

A tall evergreen with sharp needles. Stays green all winter even under heavy snow.

Why can trees grow in the boreal forest but NOT in the tundra? What's different about the ground?

The Four-Season Middle

TEMPERATE PLAINS AND FORESTS

Most of the United States

There are four seasons. The eastern part has deciduous trees (oaks and maples that drop their leaves in the fall). The middle has the Great Plains, where grasses and wheat fields stretch for miles.

Key Plant: Maple Tree

A leafy tree that turns bright red and orange in the fall, then drops its leaves before winter.

INSERT IMAGE

Photo of a temperate forest in fall -- maple trees with red/orange leaves, or a wide view of the Great Plains.

*Source: NPS, USGS, or
Wikimedia Commons*

SOCRATIC QUESTIONS

- *What happens to maple trees in the fall? Why?*
- *How is this climate different from the Arctic or the boreal forest?*
- *Why is this zone good for farming?*

The Warm South

DESERT

Southwestern U.S. and northern Mexico

Very hot and very dry. Cacti like the giant saguaro can grow here because they store water inside their thick stems.

Key Plant: Saguaro Cactus

A tall green cactus that can store water inside its thick, ribbed stem. Can live 150+ years.

TROPICAL

Southern Mexico, Central America, Caribbean

Warm year-round, with heavy rains. Rainforests, palm trees, and bright tropical flowers grow here.

Key Plants: Coconut Palm, Banana

Coconut palm: tall tree with long fronds, loves warm weather and salty sea air. Banana plant: huge green leaves, needs warm weather all year and lots of rain.

Both zones are warm -- so why does the desert grow cacti while the tropical zone grows palm trees? What's different?

INSERT IMAGE

*Climate or biome map of North America
and the Caribbean showing five major zones
in different colors with a legend:*

*Arctic/Tundra (far north)
Boreal/Taiga Forest
Temperate Forest and Grassland (middle)
Desert (southwest)
Tropical (south and Caribbean)*

*Source: NPS, USGS, NOAA, National Geographic
MapMaker, or Wikimedia Commons (public domain)*

Save as: source-3-06-04-climate-map-north-america-caribbean.jpg

SOCRATIC QUESTIONS

- *How many different climate zones can you count on the map?*
- *Which color covers the most space?*
- *Find the cold zone in the far north. Now find the tropical zone in the south. What's in between?*

WHY DOES IT MATTER?

Why is it useful to know about climate zones if you're a person planning where to plant a farm or build a town?

Where Does This Plant Grow?

Call out the climate zone for each plant. Explain how you know.

Saguaro Cactus

DESERT

Stores water in thick stems

Coconut Palm

TROPICAL

Loves warm weather and salty sea air

Spruce Tree

BOREAL

Keeps needles through snowy winters

Arctic Moss

ARCTIC TUNDRA

Survives on rocks and frozen ground

Maple Tree

TEMPERATE

Drops leaves in fall, four seasons

Banana Plant

TROPICAL

Needs warm weather all year, lots of rain

Teacher: Reveal zone answers one at a time. Ask 'How do you know?' before confirming.

The Giant Saguaro

The giant saguaro cactus can live for more than 150 years and grow over 40 feet tall -- but only in the Sonoran Desert in Arizona and northern Mexico.

It can't survive a hard freeze, so it stays where winters are mild.

INSERT IMAGE

Photo of a giant saguaro cactus in the Sonoran Desert, Arizona. Showing its tall ribbed trunk with upward-curving arms.

Source: NPS, USGS, or Wikimedia Commons (public domain)

THEN AND NOW

Long before there were countries or borders, people who lived in each climate zone learned to use the plants around them -- pine bark for boats in the boreal forest, agave for food in the desert, palm fronds for roofs in the Caribbean.

Today, much of that knowledge is still used. Climate still shapes how people live in each part of the region.

If You Tell Me the Plants, I Can Guess the Climate

Arctic Tundra

Moss

Frozen ground, no trees

Boreal Forest

Spruce

Snowy, evergreen
needles

Temperate

Maple

Four seasons, leaves
drop

Desert

Saguaro

Hot, dry, stores water

Tropical

Coconut Palm

Warm year-round, rain

EXHIBIT LABEL

North America and the Caribbean holds five different climate zones -- from the icy Arctic Tundra in the far north to the warm Tropical zone in the south. Each climate grows different plants. Climate is one of the biggest reasons one part of the region looks so different from another.

Climate Shapes How a Place Looks and Feels

All five climate zones are in the same region. But a palm tree could never survive in the Arctic, and a polar bear could never live in the Caribbean.

Climate decides what grows, what lives there, and what the land looks like. If you know the climate, you can picture the place.

SOCRATIC QUESTIONS

- *If someone told you 'It's covered in snow and only moss grows there,' what climate zone is it?*
- *If someone said 'Tall cacti and dry sandy ground,' which zone?*
- *Why is climate one of the biggest reasons the same region can look so different?*

What does the word climate mean?

A. The weather outside right now.

B. The usual weather of a place over many years.

C. The temperature in just one season.

Climate is the long-term pattern of weather in a place -- not just today.

Where would you find tall evergreen trees like spruce and pine?

A. The desert in northern Mexico.

B. The tropical Caribbean islands.

C. The boreal forest across Canada.

Evergreens like spruce and pine fill the boreal forest -- they keep their needles through long, snowy winters.

Why does the same region have so many different climates?

A. The region stretches a very long way from north to south.

B. Every country chooses its own climate.

C. All four climates are exactly the same temperature.

When a region reaches from the Arctic in the north all the way to the warm Caribbean in the south, it crosses many climate zones along the way.

Wait -- Is the Desert "Empty"?

COMMON MISTAKE

"The desert doesn't have any plants because it's too hot and dry."

Students sometimes think deserts have nothing growing at all.

CORRECTED THINKING

The desert IS hot and dry -- but special plants like the saguaro cactus have adapted. They store water inside their thick stems. The desert has life; it just looks different.

Every climate zone supports life.

SOCRATIC QUESTIONS

- *What would you say to someone who calls the desert 'empty'?*
- *How did the saguaro cactus solve the problem of having very little rain?*
- *Can you think of another zone where plants adapted in a special way?*

Scaffolding: Evidence Questions 1 and 2

EVIDENCE Q1

Why don't tall trees grow in the Arctic Tundra? Use one detail from the lesson.

SENTENCE STARTERS

Tall trees don't grow in the Arctic Tundra because...

The lesson says the ground...

Without deep roots, trees can't...

EVIDENCE Q2

Name one plant from the desert and one plant from the tropical zone. Explain how each plant is suited to its climate.

SENTENCE STARTERS

In the desert, the _____ is suited to its climate because...

In the tropical zone, the _____ is suited because...

MODEL (Q1)

Tall trees don't grow in the Arctic Tundra because the ground stays frozen so deep that roots can't reach down. The lesson says only small grasses, moss, and tough little shrubs can survive there.

Scaffolding: Q3 and Climate Panel Comparison

EVIDENCE Q3

How does climate help shape how a place looks? Use evidence from today's lesson.

Climate shapes what _____ grow in a place...

For example, in the _____ zone...

But in the _____ zone...

This shows that climate...

CURATOR TASK

Climate Panel Comparison: Pick two climate zones. Write a 3-4 sentence comparison.

YOUR LABEL SHOULD ANSWER

Which two zones?

Where are they?

What grows in each?

Biggest difference?

CURATOR'S CHRONICLE

In this exhibit, I learned that _____ because _____.

CHAT FRAME

"Pick any two zones that interest you."

"Compare where they are, what the weather is like, and what grows there."

"End with one sentence about the biggest difference."

What You'll Turn In

EVIDENCE Q1

Why don't tall trees grow in the Arctic Tundra? Use one detail from the lesson.

EVIDENCE Q2

Name one plant from the desert and one plant from the tropical zone. Explain how each plant is suited to its climate.

EVIDENCE Q3

How does climate help shape how a place looks? Use evidence from today's lesson.

CURATOR TASK

Climate Panel Comparison: 3-4 sentences comparing two zones.

CHRONICLE

In this exhibit, I learned that _____ because _____.

Great Work, Regional Curator!

You walked all five climate panels, matched plants to their zones, and wrote a comparison that helps visitors see why this region can look so different from one part to another.

COMING NEXT

Natural Resources and How People Use Them

A new shelf is being unpacked at the side of the gallery -- the Natural Resources shelf. It's stocked with water, soil, forests, minerals, and crops. Next time, you'll figure out how people across the region actually use the land -- not just look at it. The story starts turning toward choices, and that means honesty becomes important.